

Amrutha Ajish Achuthan

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📍 Pathanamthitta, Kerala

PROFESSIONAL SUMMARY

Computer Science undergraduate with Honors in Machine Learning and a Minor in Computational Mathematics. Research experience in anomaly detection, statistical modeling, and high-dimensional data analysis. Interested in Machine Learning, Data Science, and AI systems.

RESEARCH INTERESTS

Statistical Machine Learning | Anomaly Detection | Data Mining | AI Security | High-Dimensional Data Analysis

EDUCATION

Saintgits College of Engineering Aug. 2023 – May 2027 (Expected)
B.Tech in Computer Science and Engineering; CGPA: 9.33 / 10 Kottayam, Kerala

- **Specialization:** Honors in Machine Learning | Minor in Computational Mathematics
- **Relevant Coursework:** Probability Theory, Linear Algebra, Statistical Inference, Optimization Techniques, Data Structures & Algorithms, Machine Learning.

EXPERIENCE

IIIT Kottayam May 2025 – June 2025
Research Intern, Cybersecurity Kottayam, Kerala

- Developed a Behavioral Fingerprinting system using statistical feature extraction to model network traffic patterns from high-dimensional datasets.
- Implemented unsupervised statistical learning models including Isolation Forest (**97% detection accuracy**) and K-Means clustering; analyzed bias-variance trade-offs and anomaly score distributions.
- Validated model thresholds using hypothesis-driven evaluation with confusion matrix, precision-recall metrics, and statistical performance comparison.

Google May 2026 – Sept. 2026
Google Gemini Student Ambassador Remote

- Selected among 4,500 students from all over India to serve as a campus ambassador and drive awareness of Google Gemini tools.

Infosys Springboard Feb. 2026 – Present
Data Analytics Intern Remote

- Selected for the specialized Data Analytics track; undergoing advanced training in EDA, dynamic visualization, and statistical predictive modeling techniques.

QNAYDS LLP July 2024
Intern, Cyber Security Remote

- Assisted in identifying network vulnerabilities and mapping security frameworks through structured, data-driven auditing methodologies.

PROJECTS

- Verity (The Truth) | Python, Microsoft Presidio, Local LLMs, Security API

GitHub

 - Architected a Sandboxed AI Gateway utilizing **Microsoft Presidio** and **Local LLMs** for policy-based logic and regex-driven PII redaction to secure enterprise AI adoption.
 - Conducted rigorous exploratory data analysis (EDA) and statistical pattern analysis to precisely model sensitive data leakage probabilities across interaction scenarios.
- UMAC – Unified Military Analytics & Comparison | Python, Power BI, BeautifulSoup, Pandas

GitHub

 - Engineered a full-stack analytics pipeline to scrape, clean, and extract KPIs from live defense data spanning **140+ countries and 50+ indicators**.
 - Designed interactive **Power BI dashboards** to enable country-level capability comparison and coalition simulation.
- RakshakCV – Automated Compliance Detection | Python, YOLO Nano, YOLOv8, OpenCV

GitHub

 - Developed a real-time computer vision pipeline utilizing **YOLO Nano and YOLOv8 architectures** to accurately detect helmet compliance and extract actionable data.
 - Optimized deep learning model inference for rapid visual monitoring, drastically reducing manual oversight requirements.
- E-Waste Data Security (UST SIGHT 2.0 Finalist) | IoT, Cryptography, Mobile App

 - Designed an EMP generator hardware prototype for physical data destruction, verifying operational security via automated cryptographic integrity checks.
- EcoLogical – Smart Household Carbon Accountability | React, Django, PostgreSQL, REST API

GitHub

 - Built a platform that calculates real-time CO2 emissions using standardized factors, featuring a gamified eco-action tracking module and carbon credit marketplace.
- Digital Leave & Food Management System | Python, MySQL, JavaScript

 - Developed a full-stack portal with a demand-forecasting “Food Opt-In/Out” feature that significantly minimized administrative overhead and food wastage.

ACHIEVEMENTS & CERTIFICATIONS

- **Certifications & Tests:** McKinsey Forward Program (2026), GATE 2026 Qualified, IELTS Academic (Band 8.0), Deloitte Data Analytics Job Simulation, NPTEL Elite+Silver (Deep Learning - 81%, IoT & ML).
- **Technical Competitions:** 1st Prize Data Visualization Challenge (XIM Univ.), Finalist Saintgits Coding Championship 2025, 2nd Prize Exponentia Maths Quiz, Rank 1 College CTF Leaderboard.

LEADERSHIP

- ACM Student Chapter

Treasurer

Oct. 2024 – Present
Saintgits College of Engineering

 - Spearheaded financial planning and budget allocation for chapter activities; coordinated 'ACM Week' logistics for **200+ participants**.
- College Math Club (Infinity)

Execom Member

Sept. 2024 – Present
Saintgits College of Engineering

 - Organized competitive mathematics quizzes and mentored junior students in Calculus, Probability Theory, and advanced statistical problem-solving techniques.

SKILLS

Languages : Python (**NumPy**, Pandas, Scikit-Learn), SQL, Java, C, HTML/CSS, JavaScript
Tools : Git, Matplotlib, Kali Linux, Seaborn, Wireshark, CICFlowMeter, Vercel
Concepts : Probability Theory, Statistical Inference, Linear Algebra, Optimization, Regression Modeling, Hypothesis Testing, Machine Learning, Data Modeling, Anomaly Detection, DSA
Soft Skills : Leadership, Quantitative Problem-solving, Adaptability, Public Speaking, Team Management